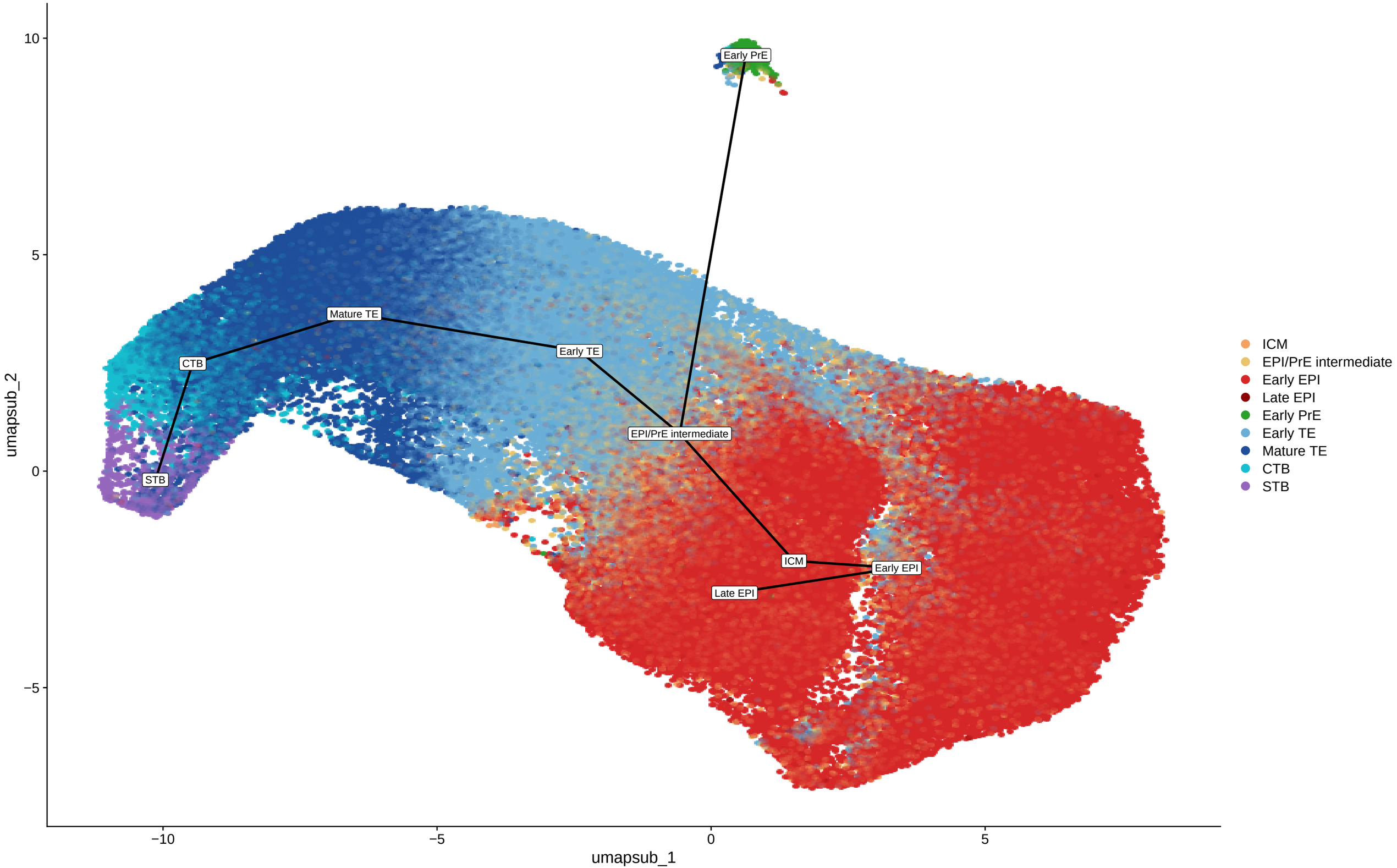
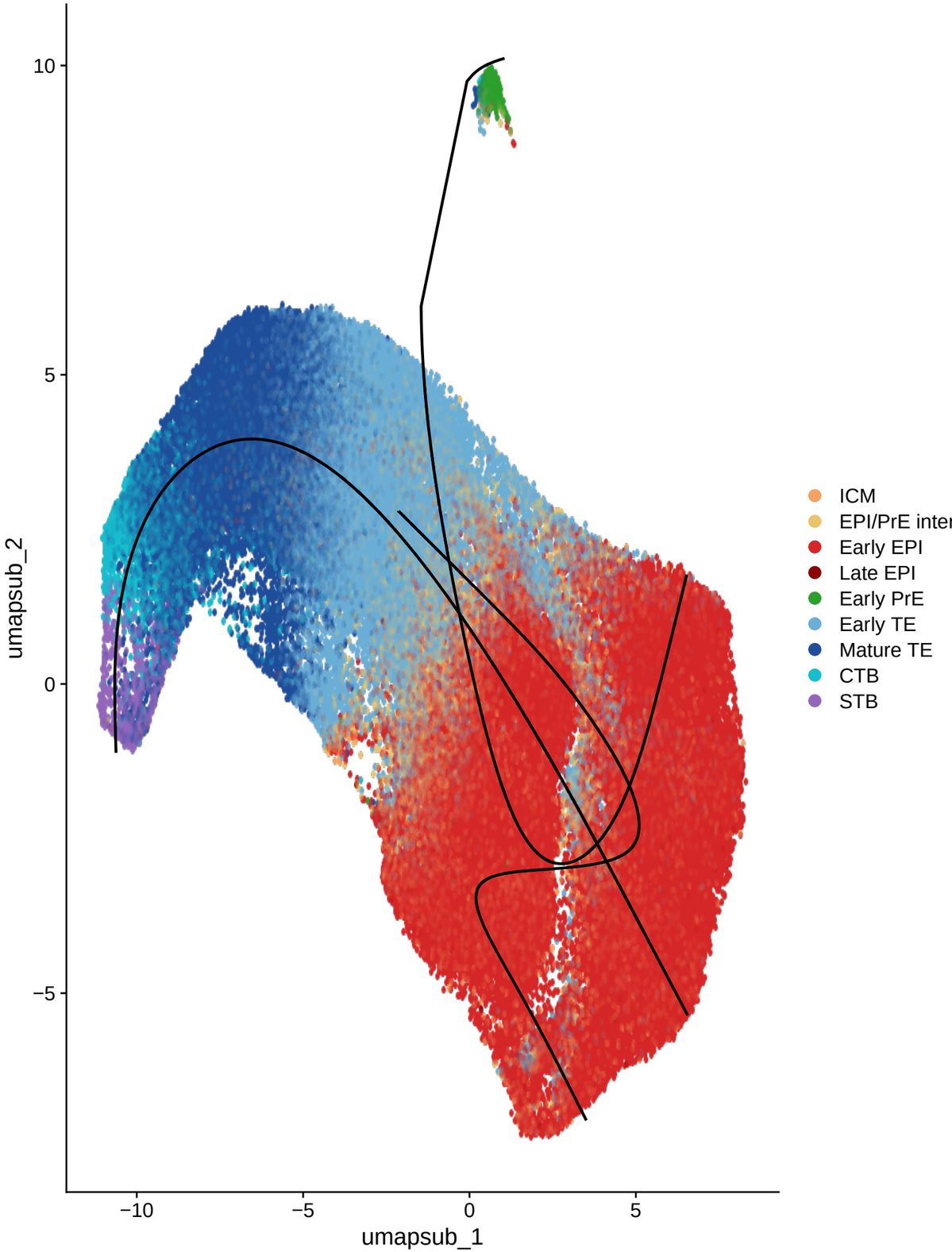
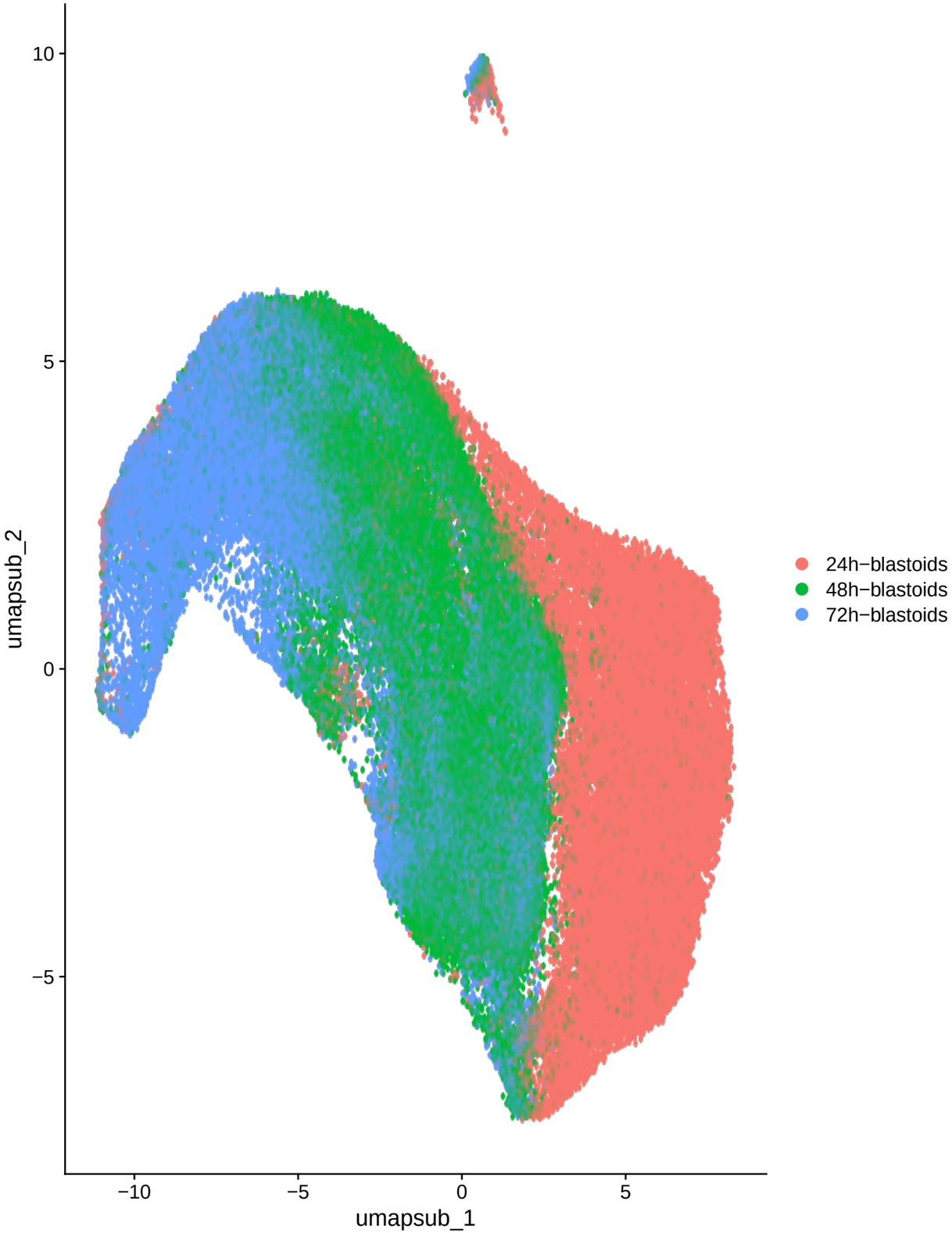
Slingshot tree between SingleR labels (start: Early EPI)



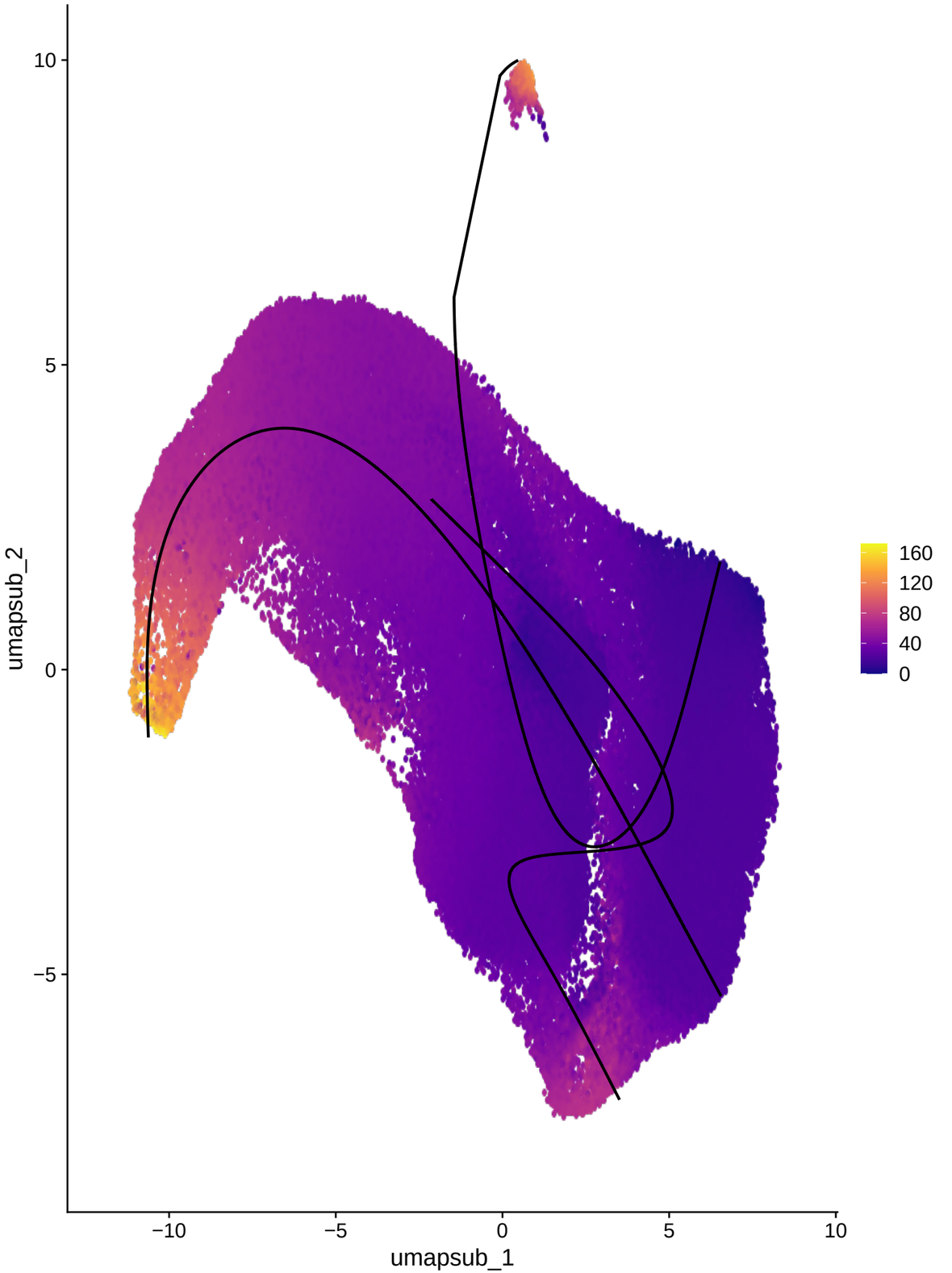
Slingshot curves on the UMAP of these cells



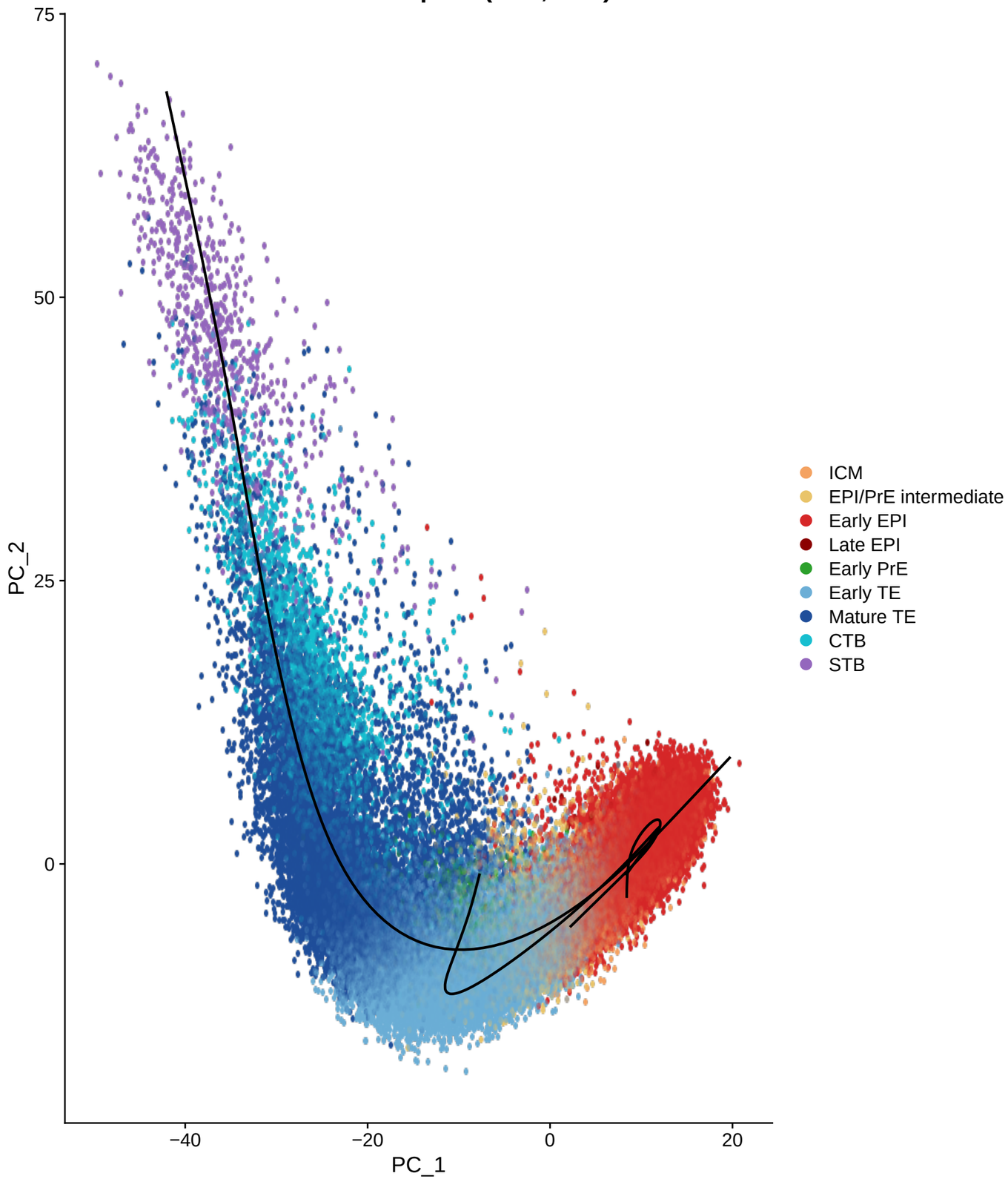
Sample



Average pseudotime



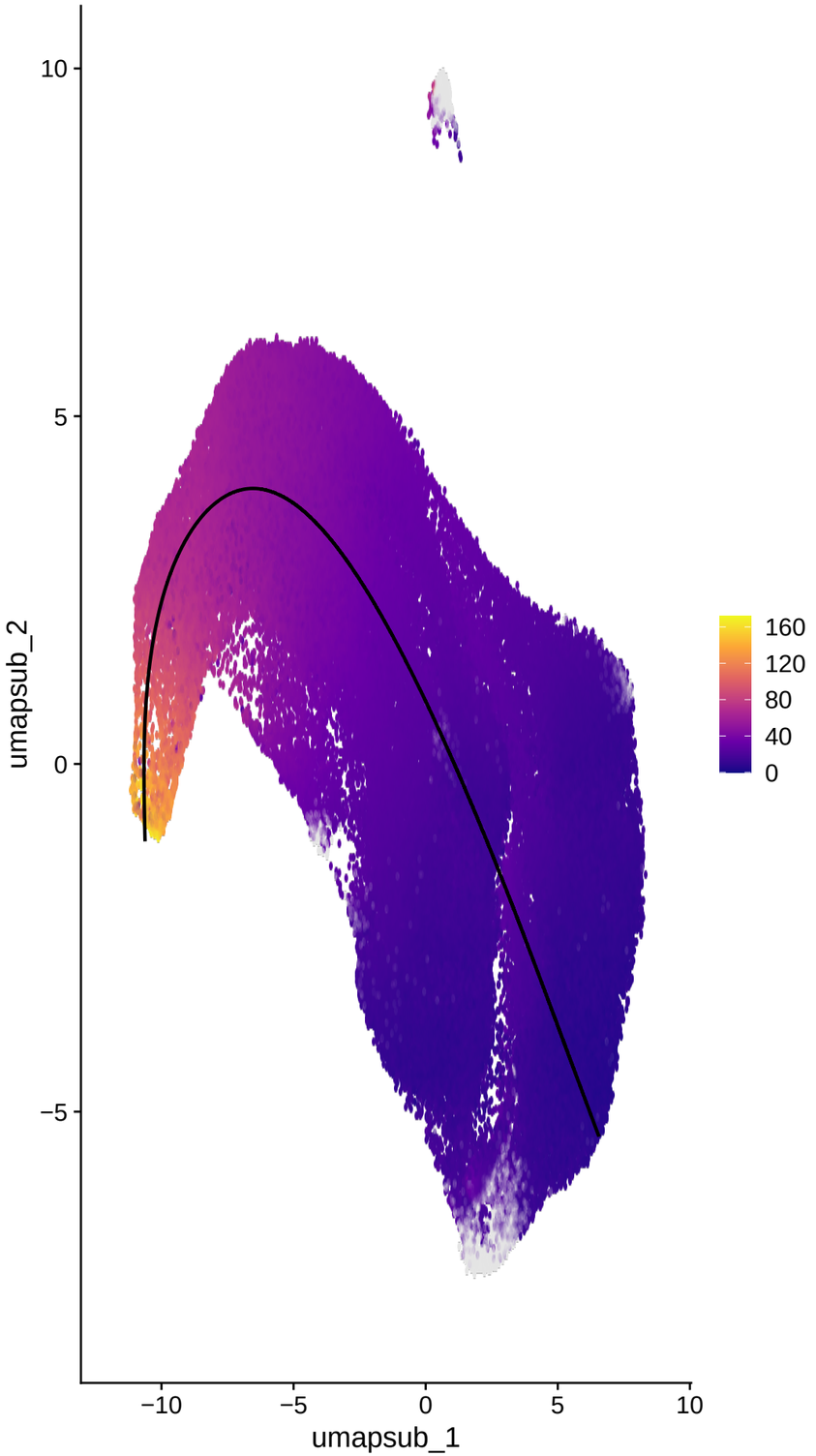
Curves in PCA space (PC1, PC2)



Pseudotime per lineage (grey = not on the lineage)

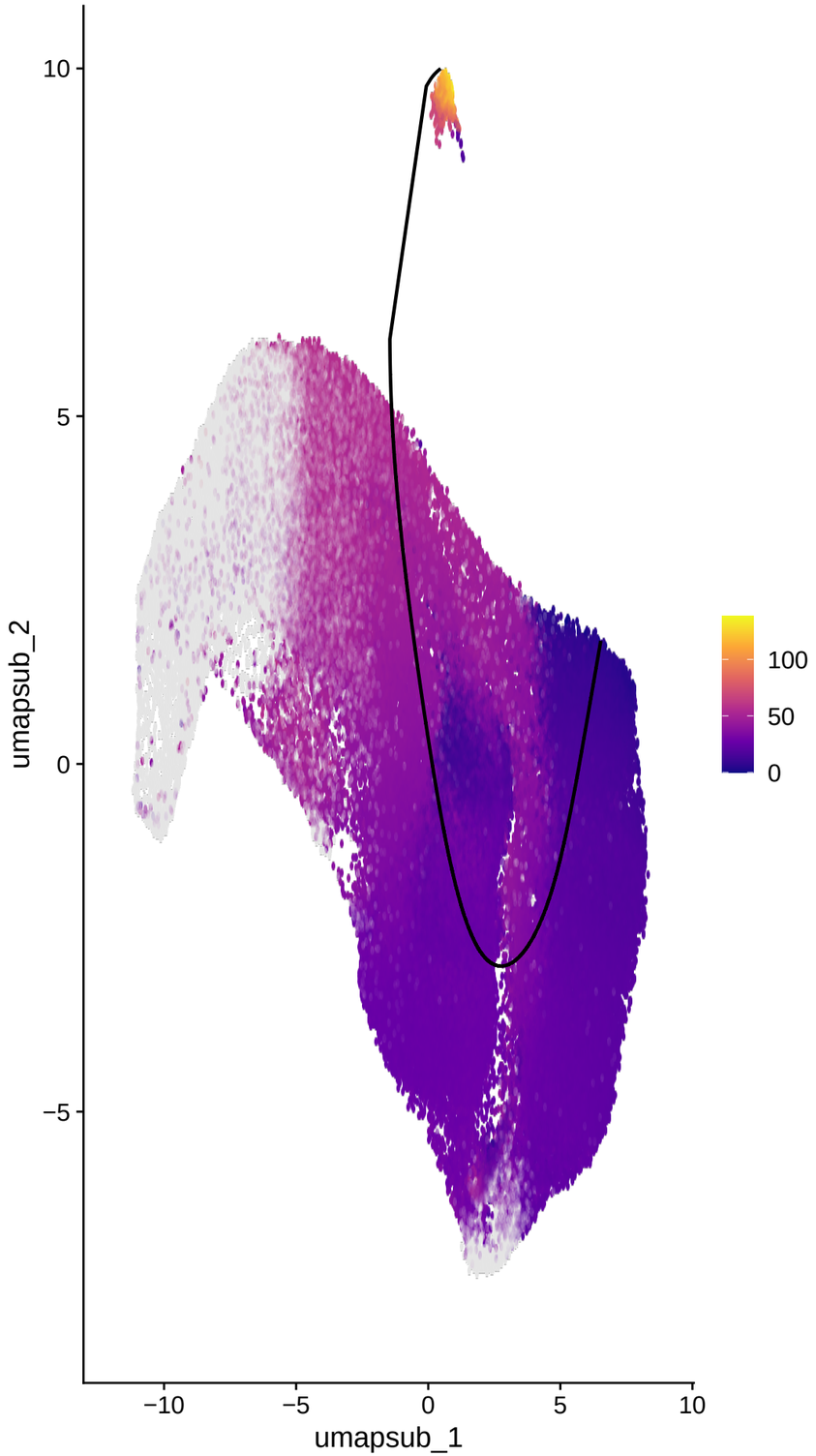
Lineage1

Early EPI > ICM > EPI/PrE intermediate > Early TE > Mature TE > CTB > STB



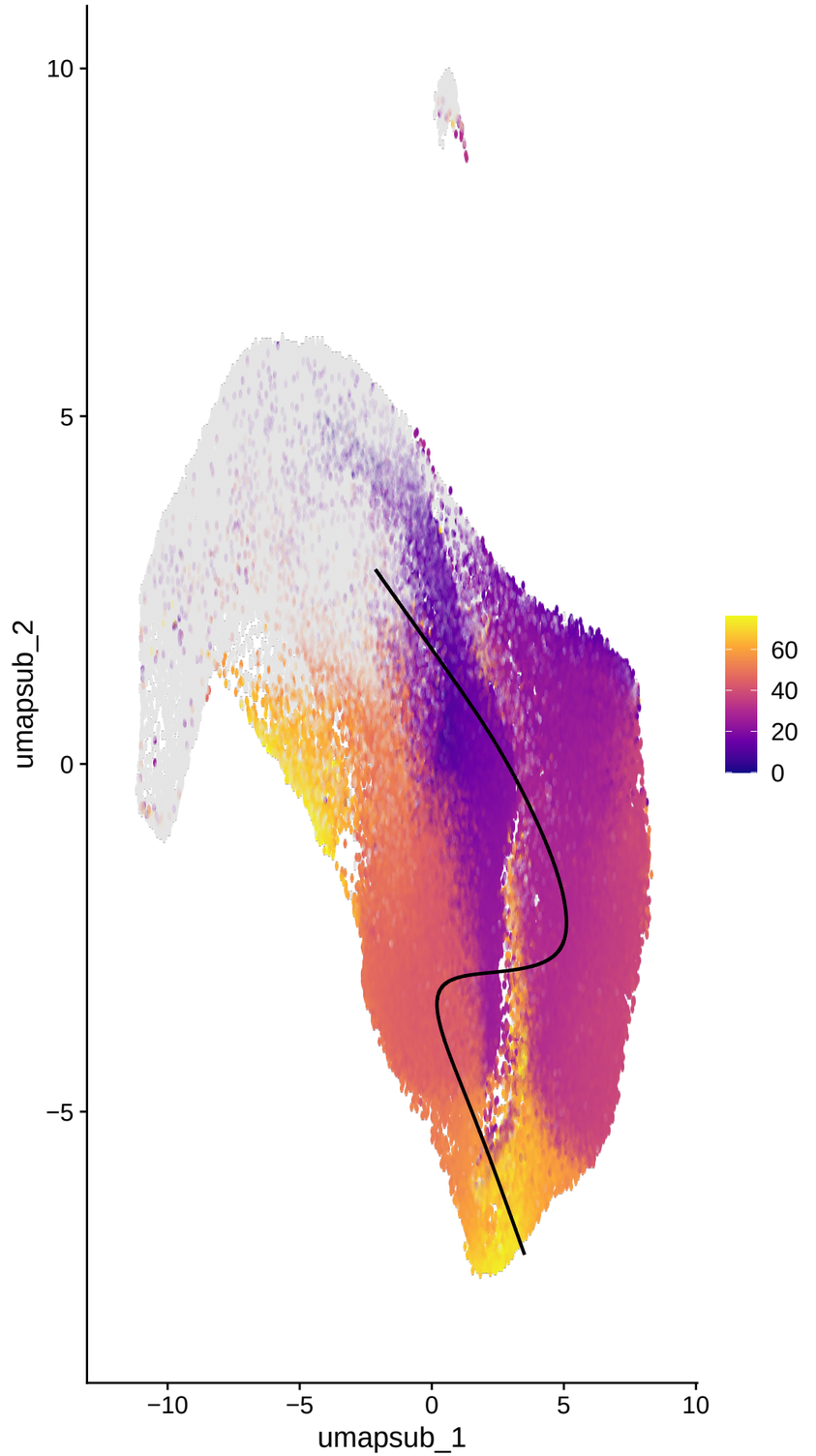
Lineage2

Early EPI > ICM > EPI/PrE intermediate > Early PrE

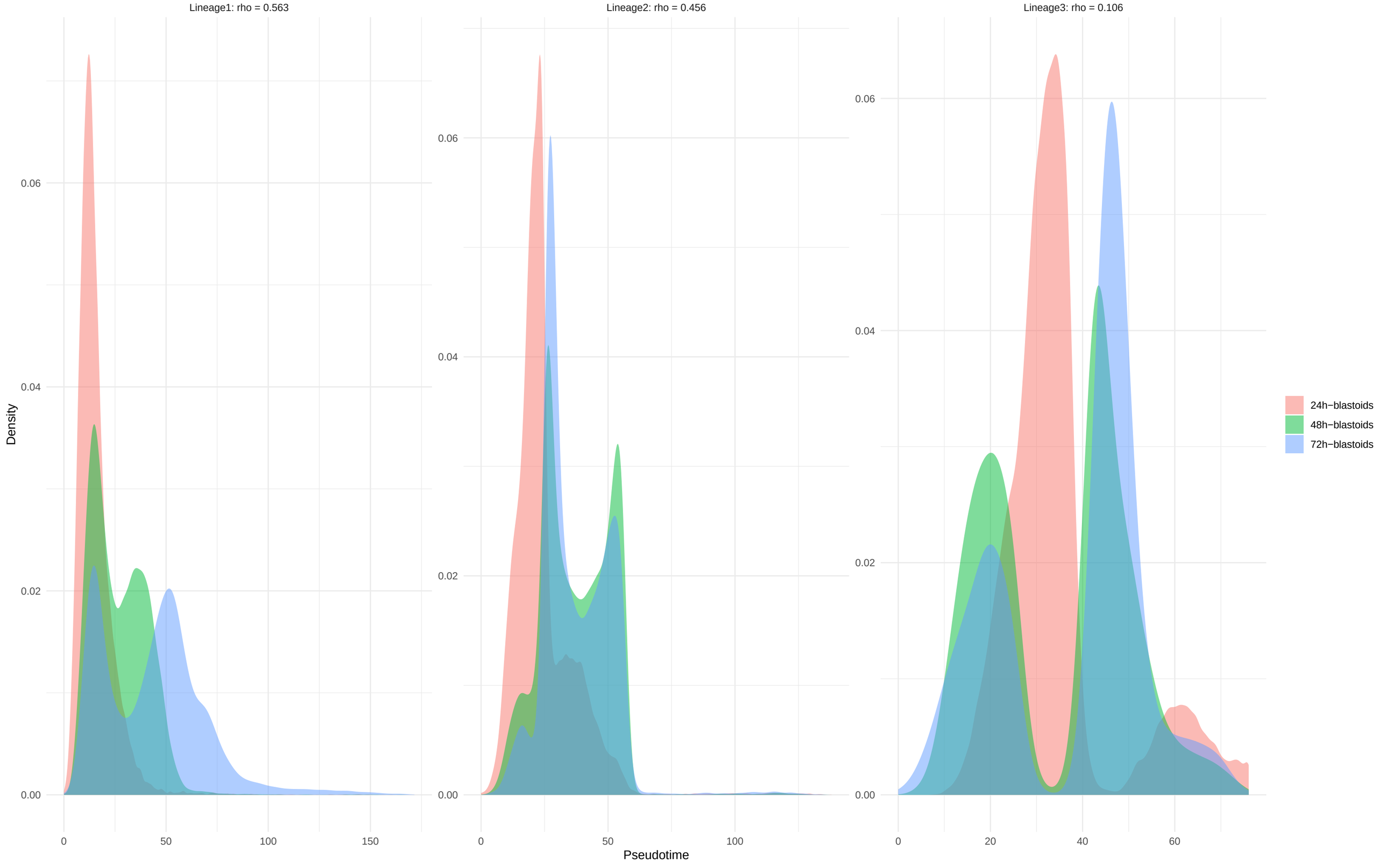


Lineage3

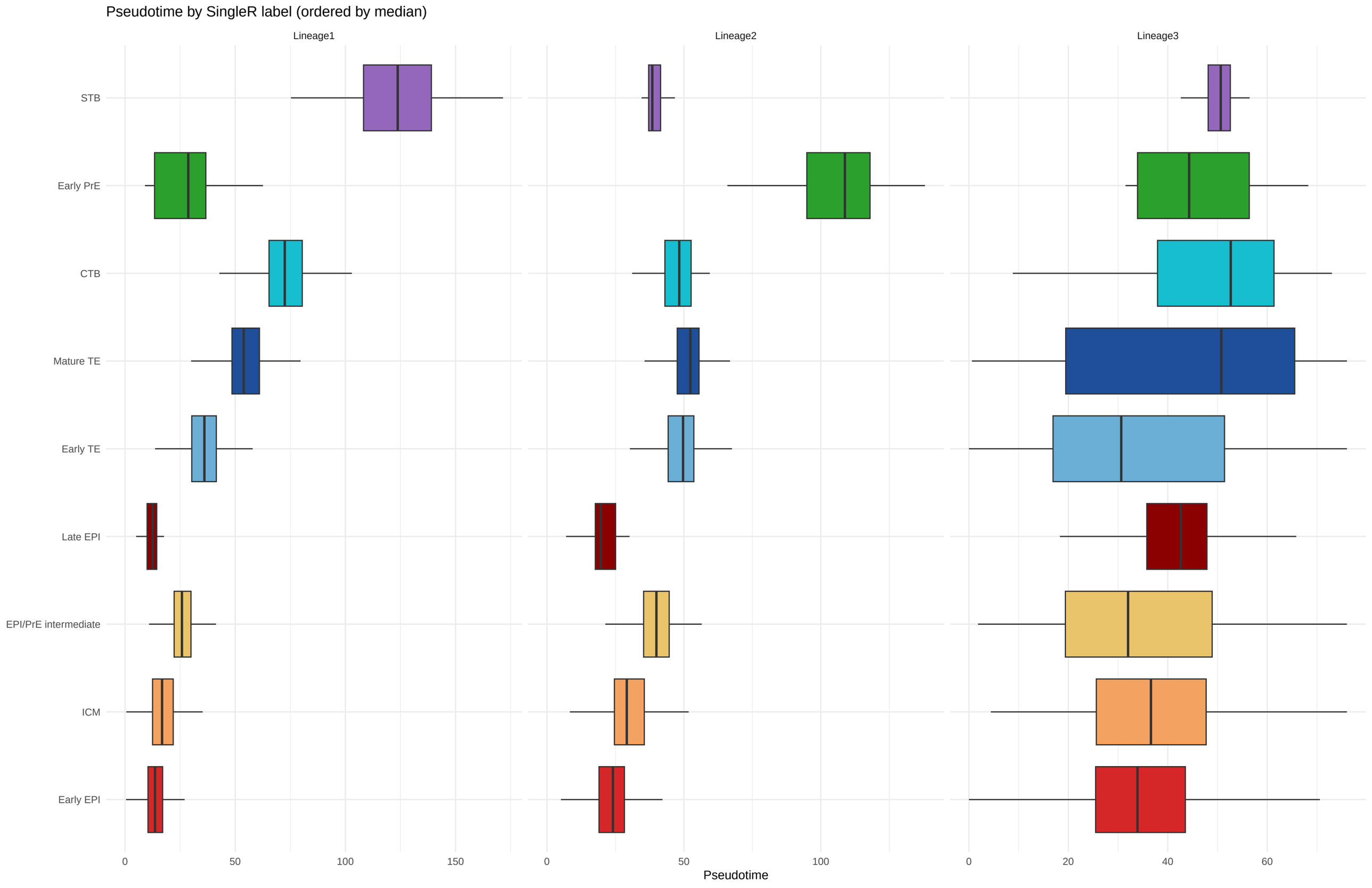
Early EPI > Late EPI



Pseudotime by sample (rho = Spearman correlation with sample time)



Pseudotime by SingleR label (ordered by median)



Marker genes along pseudotime (GAM fit, log−normalised expression)

POU5F1
NANOG
SOX2
KLF17
DPPA3
GATA3
GATA2
TFAP2C
KRT18
CDX2
ESRRB
ENPEP
NR2F2
CGA
GCM1
VGLL1
OVOL1
TP63
ELF5
TEAD4
CGB3
CGB5
SDC1
CYP19A1
HSD3B1
GATA6
SOX17
GATA4
PDGFRA

Expression
Pseudotime

lineage
Lineage1
Lineage2
Lineage3

Zhao et al. 2024 top 25 markers per cell type: mean z−scored trend along each lineage

Lineage1: Early EPI > ICM > EPI/PrE intermediate > Early TE > Mature TE > CTB > STB

Lineage2: Early EPI > ICM > EPI/PrE intermediate > Early PrE

Lineage3: Early EPI > Late EPI

Mean z−score of the cell type's markers

3
2
1
0
−1

0
50
100
150

0
50
100

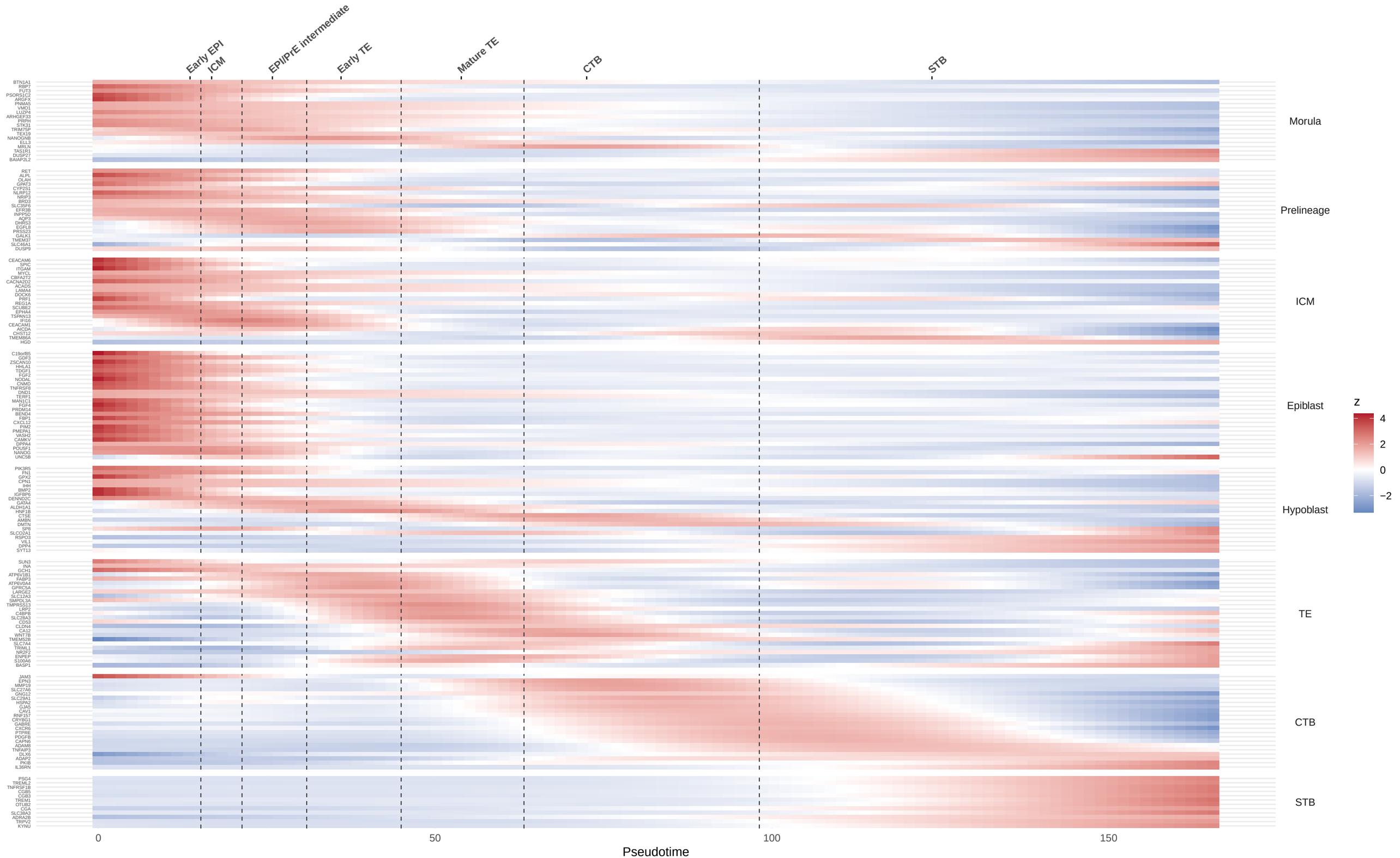
0
20
40
60

Pseudotime

Zhao cell type
Morula
Prelineage
ICM
Epiblast
Hypoblast
TE
CTB
STB

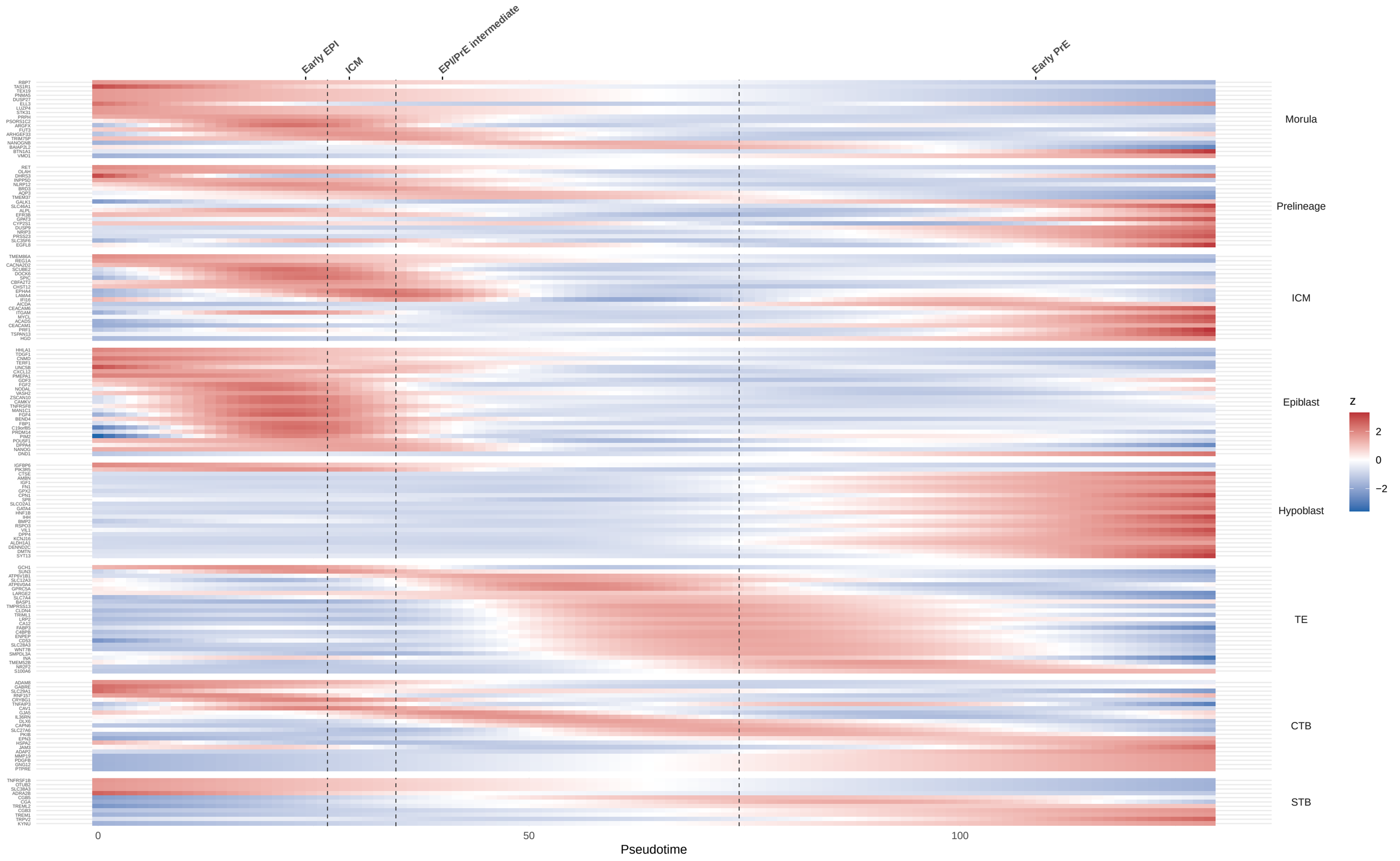
Lineage1: Zhao et al. 2024 top 25 markers per cell type along pseudotime (z−score per gene)

Early EPI > ICM > EPI/PrE intermediate > Early TE > Mature TE > CTB > STB



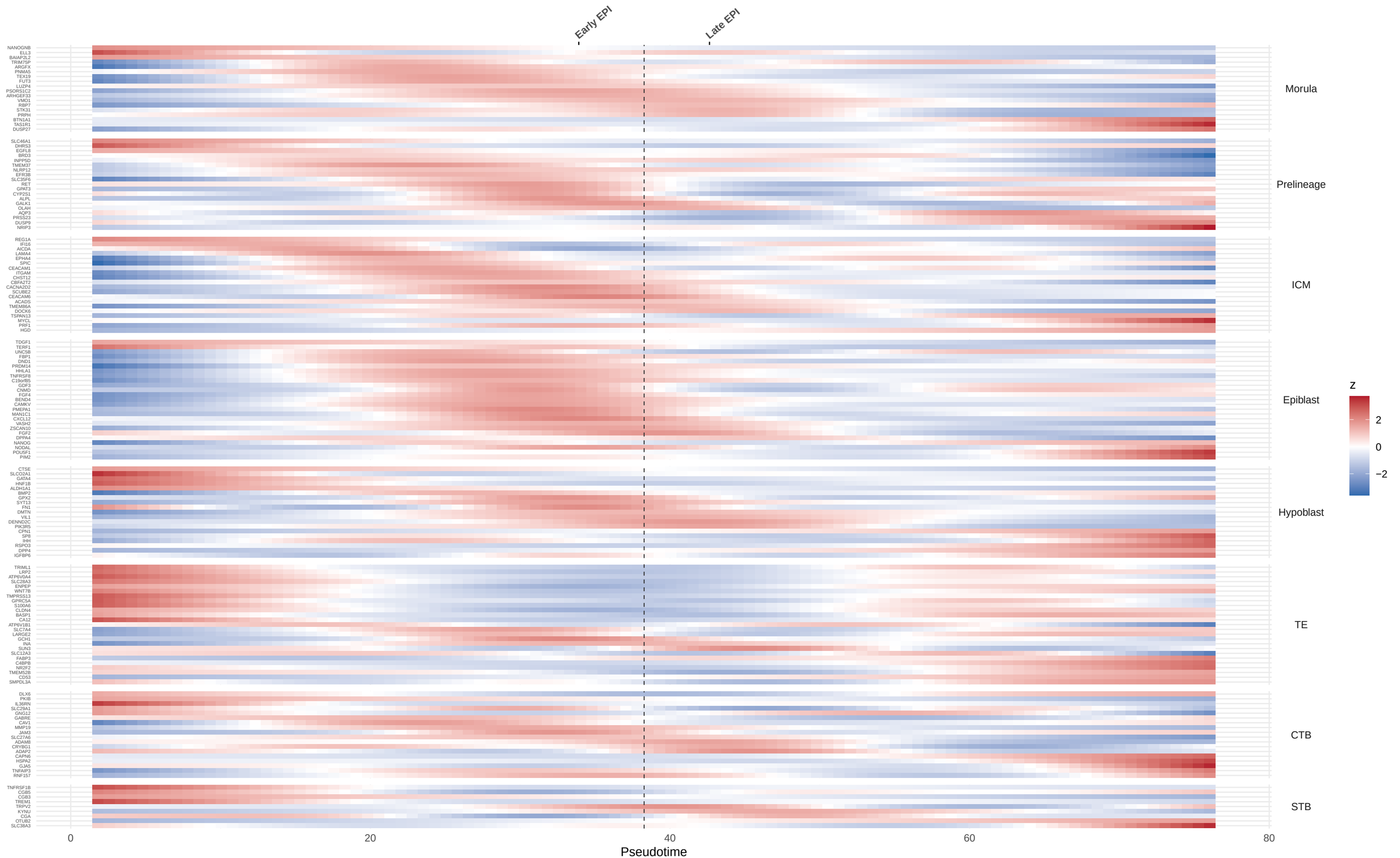
Lineage2: Zhao et al. 2024 top 25 markers per cell type along pseudotime (z−score per gene)

Early EPI > ICM > EPI/PrE intermediate > Early PrE



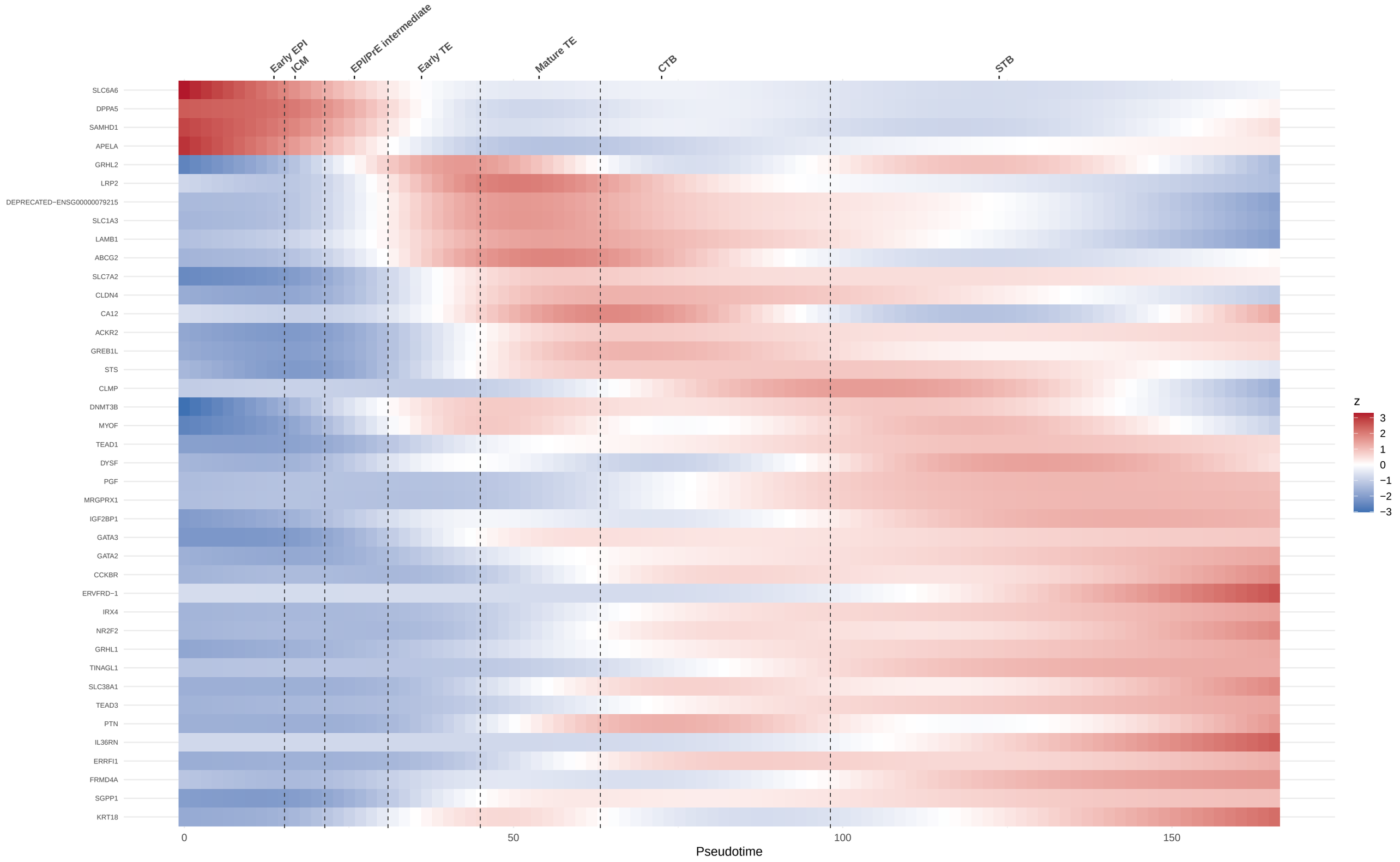
Lineage3: Zhao et al. 2024 top 25 markers per cell type along pseudotime (z−score per gene)

Early EPI > Late EPI



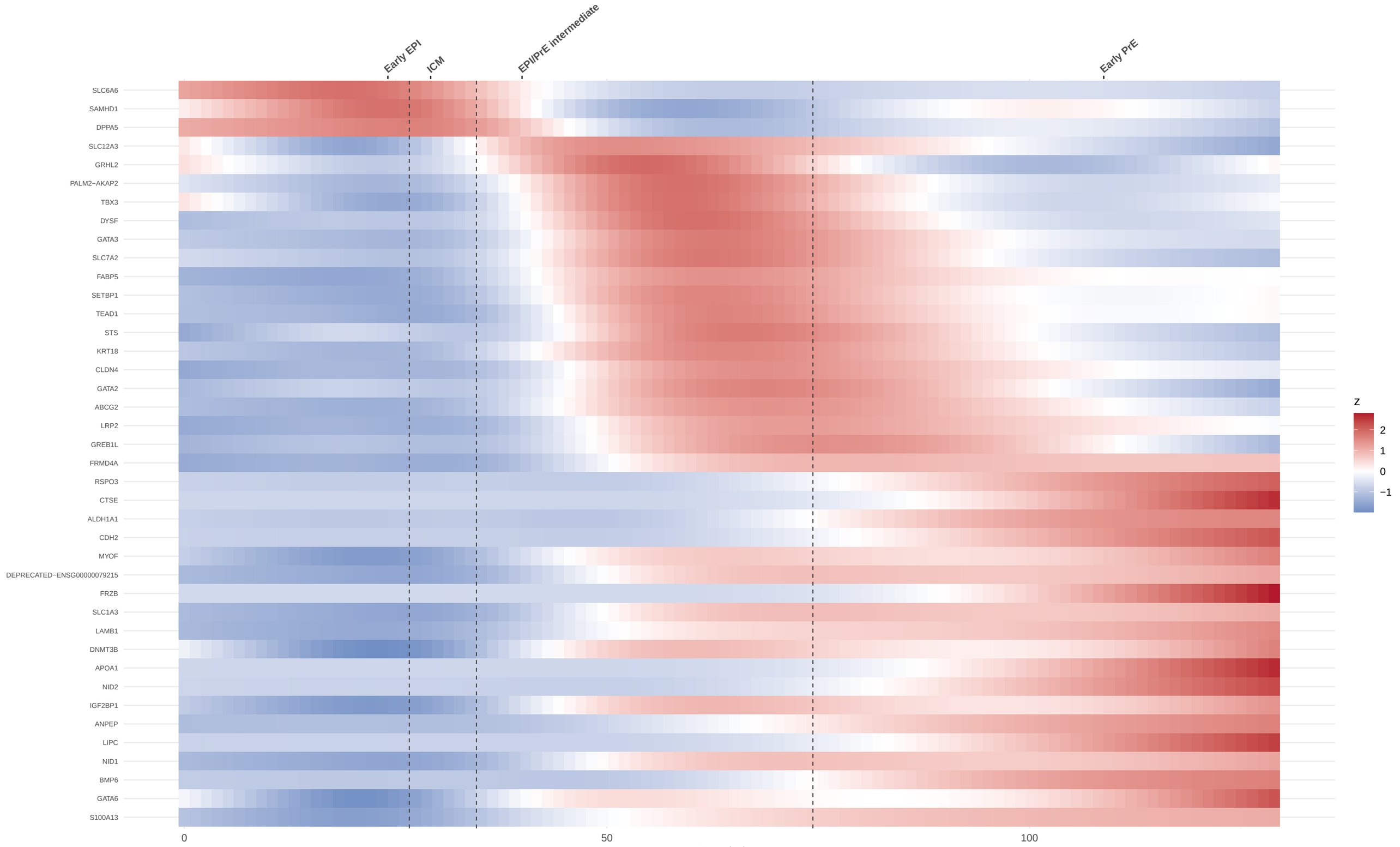
Lineage1: top 40 genes changing along pseudotime (ordered by peak)

Early EPI > ICM > EPI/PrE intermediate > Early TE > Mature TE > CTB > STB



Lineage2: top 40 genes changing along pseudotime (ordered by peak)

Early EPI > ICM > EPI/PrE intermediate > Early PrE



Lineage3: top 40 genes changing along pseudotime (ordered by peak)

Early EPI > Late EPI

